

Science PRI - Laws and First Weld

PRI as architecture, not a vibe. Plain first, then formal labels.

Source: [gate/science_pri.py](#) | Undefined word? Open doc 18 (Layer 0 atoms) and define it before continuing.

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Plain opener

Bad outcomes often happen when a system finishes a one-way action without enough justification. PRI means: build the stop into the architecture. If it is not LIVE and justified, DENY. If something breaks while checking, DENY. Turning the path back on takes CHARGE - not a friendly admin click.

First commercial driver

ID: withdraw_payout_clear

Write: withdraw / payout - clear before wire

Why first: Money leaving is irreversible enough to prove the mouth, commercial enough to sell, before grid or program cosplay.

Example path: POST /v1/payouts/{id}/release

Checkout: /operator?write=withdraw

Next after prove: grid_shed_reconnect - not first

Formal laws (after atoms)

PRI: execute(a) only if LIVE(a) AND justified(a); else DENY. bypass empty. CHARGE-only re-open.

Fail-closed: uncertainty => DENY. Life-safety overrides are explicit coordinator lanes, not soft-yes.

Intervention cost: Real control is measure -> LIVE|DENY -> receipt. Silent free yes is illegal. Price the dissipation (weld + bps).

Driver on the act graph: Weld one irreversible edge at a time. Hub dashboards are not drivers.

Quorum mouth: LIVE needs quorum(policy) AND CHARGE AND Gate mouth. Stranger-verify outside the actor's forge zone.

Not first

- Grid blackstart (step 2)
- Milcomms release mouth (ladder later)
- Nuclear C2 (state only)

Retrieval drill

- State PRI as IF/ELSE DENY.
- Why payout before grid?
- What alone re-opens DEAD?